Risk-Calibrated Adaptive Stopping for Diffusion Language Models

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Abstract

Diffusion language models expose a useful but hazardous degree of freedom: a decoder can stop refining a token block early, reducing neural function evaluations (NFEs), but a mistimed stop may irreversibly change the generated continuation. Existing adaptive decoders use confidence, entropy, token stability, or learned schedules, yet generally do not control this failure probability. We introduce Risk-Calibrated Phase-Adaptive Generation (RC-PAG), which learns a lightweight risk score from compact online features and calibrates a predeclared family of stopping policies with exact binomial tests. For a fixed model, prompt distribution, and decoding protocol, the selected policy controls the probability that any early-committed block disagrees with its full-refinement continuation, up to a user-selected risk level, with finite-sample familywise confidence. Calibration labels use same-state shadow continuations only; deployment adds neither a second model nor shadow computation. We define a preregistered two-model evaluation and a compute-aware experimental funnel designed to distinguish genuine scheduling gains from accuracy loss. This draft is wired to insert confirmatory tables only after non-mock evidence passes the protocol's coverage checks; numerical claims are intentionally deferred until that run.

1 Introduction

Masked diffusion language models (dLMs) generate many tokens in parallel by repeatedly denoising a partially masked sequence (Nie et al., 2025; Ye et al., 2025). Their wall-clock advantage depends on when tokens or blocks are committed. Fixed schedules spend refinement steps uniformly; recent adaptive methods vary the block length, token transfer, or stopping time using confidence and stability signals (Lu et al., 2026; Wu et al., 2025; Israel et al., 2025; Mohamed et al., 2025).

The central difficulty is that confidence is not itself a safety guarantee. A scheduler chosen because it looks efficient on a tuning set can silently over-commit on new prompts, and searching many thresholds amplifies this selection bias. We therefore ask: *Can an adaptive dLM decoder minimize refinement compute while controlling the probability of changing a full-refinement block?*

Our answer, RC-PAG, separates prediction from certification. A small CPU estimator predicts the risk of stopping from statistics already available in the denoising loop. A finite, predeclared policy family turns those scores into stopping rules. On a held-out calibration split, the actual on-policy decisions are compared with full-refinement continuations cloned from exactly the same decoder states. Exact one-sided binomial tests and a Bonferroni correction certify only policies whose prompt-level disagreement risk is below α; otherwise the system falls back to the full refinement budget.

Our contributions are: (i) a prompt-level loss that makes premature block commitment measurable without conflating it with task correctness; (ii) an online history-aware stopping policy with no deployment-time auxiliary-model forward pass; (iii) a finite-sample, selection-safe risk certificate for a predeclared policy family; and (iv) an auditable evaluation over LLaDA and Dream with paired accuracy, NFE, runtime, strict risk, out-of-domain transfer, and explicit positive-claim gates.

2 Related work

Efficient dLM decoding. LLaDA and Dream establish competitive autoregressive-style reasoning with masked diffusion objectives (Nie et al., 2025; Ye et al., 2025). Fast-dLLM uses confidence-aware parallel decoding and cache reuse (Wu et al., 2025); AdaBlock adapts block size online (Lu et al., 2026); APD learns token-wise denoising paths (Israel et al., 2025); and SchED uses similarity-based early stopping (Mohamed et al., 2025). FreeDave, OSDT, and WINO further explore lossless verification, one-shot thresholding, and revocable inference optimization (Wu and Zhang, 2026; Shen and Ro, 2026; Hong et al., 2026). DiCo and SOAR study distillation and schedule optimization, respectively (Luo et al., 2026; Cao et al., 2026). Very recent work predicts semantic block boundaries (SemBlock), combines confidence with token and distribution persistence (LESS), or coordinates simultaneous commits through a mean-field objective (Song et al., 2026; Mohamed et al., 2026; Zoabi et al., 2026). These approaches motivate a broad baseline suite, but their efficiency rules do not provide the finite-sample prompt-risk control considered here. Concurrent theory analyzes efficiency conditions for confidence decoding (Cai and Li, 2026); our guarantee instead treats the decoder as a black box and controls an observed operational loss. A contemporaneous re-evaluation also finds that confidence-based remasking gains can depend strongly on the decoding setting (Frkovic et al., 2026), reinforcing the need for paired, protocol-specific evidence.

Statistical risk control. Learn-then-Test (LTT) converts a fixed collection of predictors into risk-controlling choices using valid hypothesis tests (Angelopoulos et al., 2021); conformal risk control extends this perspective to general bounded losses (Angelopoulos et al., 2022). We apply the LTT principle to adaptive inference. Unlike ordinary post-hoc confidence intervals, simultaneous testing remains valid after selecting the lowest-compute certified policy.

3 Risk-calibrated adaptive stopping

3.1 Problem and strict shadow loss

Let $X \sim \mathcal{D}$ be a prompt. A block diffusion decoder creates blocks $b = 1, \dots, B(X)$. Within block b, $S_{b,t}$ is the complete decoder state after refinement step t, and a policy π_λ chooses a stopping time $\tau_b^\lambda \in \{1, \dots, T_b\}$. Its compute is

$$C_\lambda(X) = \sum_{b=1}^{B(X)} \tau_b^\lambda, \qquad \mathrm{NFE}(\pi_\lambda) = \mathbb{E}_{X\sim\mathcal{D}}[C_\lambda(X)]. \tag{1}$$

At every on-policy early stop, we clone S_{b,τ_b^λ} and run a *shadow* decoder to the fixed full budget T_b. Let Y_b^λ be the committed block and Y_b^{full} the shadow block. Define

$$L_\lambda(X) = \max_{b:\tau_b^\lambda < T_b} \mathbf{1}\left\{Y_b^\lambda \neq Y_b^{\mathrm{full}}\right\}, \qquad \mathcal{R}(\lambda) = \Pr_{X\sim\mathcal{D}}[L_\lambda(X) = 1], \tag{2}$$

with the maximum over an empty set equal to zero. This intentionally severe loss penalizes *any* changed token in *any* early-committed block. It detects premature commitment, not whether either answer is semantically correct. Our constrained objective is

$$\min_{\lambda\in\Lambda} \mathrm{NFE}(\pi_\lambda) \quad \text{subject to} \quad \mathcal{R}(\lambda) \leq \alpha. \tag{3}$$

3.2 Online risk features and policies

The decoder compresses each logits tensor on device and transfers only scalar summaries: masked fraction; block step and size; entropy, top-one probability, and top-one/top-two margin quantiles; token churn; changes in these quantities; delimiter and special-token rates; and a four-step history. No logits or hidden states are retained on CPU. An estimator $\widehat{r}_\theta(S_{b,t}) \in [0, 1]$ predicts the probability that stopping at that state changes the full-refinement block. Histogram gradient boosting is the primary estimator and logistic regression is a capacity ablation. Both are trained offline using full trajectories from the training split.

A local policy stops when $t \geq t_{\min}$ and $\widehat{r}_\theta(S_{b,t}) \leq q$ for p consecutive steps. The history-aware policy additionally exposes recent trends and churn to the estimator. We freeze six candidates before calibration: $q \in \{0.01, 0.025, 0.05\}$ crossed with local/history variants, all with $t_{\min} = p = 2$. Thus calibration chooses among policies; it does not tune them.

3.3 Finite-sample calibration

Let $X_1, \ldots, X_n$ be an untouched calibration sample and $K_\lambda = \sum_i L_\lambda(X_i)$. We treat every model–policy pair as a candidate, so $M = 2|\Lambda| = 12$ simultaneous hypotheses. For each pair, test $H_\lambda : \mathcal{R}(\lambda) > \alpha$ using the boundary-null lower-tail value

$$p_\lambda = \Pr_{Z \sim \mathrm{Binomial}(n,\alpha)}[Z \leq K_\lambda]. \tag{4}$$

Candidate λ is certified if $p_\lambda \leq \delta/M$. Among certified candidates we select the one with the smallest calibration mean NFE; if none is certified, we return the full-budget decoder. We also report the simultaneous one-sided Clopper–Pearson bounds.

Proposition 1 (Selection-safe prompt-risk control)**.** *Assume the calibration prompts are exchangeable draws from the deployment distribution, the model and decoding protocol are fixed, and Λ is independent of calibration outcomes. With probability at least $1 - \delta$ over calibration, every policy certified by Equation 4 has $\mathcal{R}(\lambda) \leq \alpha$. Consequently, selecting the lowest-NFE certified policy preserves this guarantee.*

Proof sketch. For any unsafe candidate with $\mathcal{R}(\lambda) > \alpha$, the lower tail of its binomial count is stochastically no larger than the boundary distribution, so its test rejects with probability at most δ/M. A union bound over the fixed family makes the probability of certifying any unsafe candidate at most δ. The final NFE-based selection is a function of this simultaneously valid certified set. A full proof appears in Appendix A.

Calibration cost and deployment cost. Shadows run only after on-policy candidate stops during calibration. Training labels are amortized from one full trajectory per prompt. At deployment, feature extraction is $O(V)$ for the already-materialized vocabulary logits and $O(d)$ retained scalars, while the CPU estimator cost is negligible relative to a model forward pass; no shadow is executed.

4 Experimental protocol

Models, data, and splits. We pin LLaDA-8B-Instruct and Dream-v0-Instruct-7B revisions and use deterministic decoding with generation length 256, at most 64 refinement steps, and block length at most 64. Estimator training, tuning, and calibration use disjoint deterministic subsets of GSM8K, MATH, and MBPP training data. Confirmation uses all 1,319 GSM8K test examples, MATH-500, 257 sanitized MBPP problems, and 164 HumanEval problems; HumanEval is declared out-of-domain. The calibration pool has 300 prompts per model and is never used for fitting or threshold search. The frozen YAML, dataset revisions, seeds, and protocol hash accompany the code.

Comparators and ablations. The confirmatory comparison includes AdaBlock, the best accuracy-eligible nonlearned rule chosen on tuning data, and the calibrated local/history RC-PAG variants. Development-only screening additionally covers fixed schedules, constant budgets, entropy, confidence and stability gates, size lookup, prior PAG and residual PAG, an offline oracle lower bound, and faithful *style reproductions* of Fast-dLLM and SchED heuristics. We label the latter as reproductions rather than official implementations. Estimator and history ablations test whether gains come from calibration alone or from temporal phase information. HGB and logistic estimators are evaluated on the same deterministic, prompt-grouped trace holdout using Brier score and AUROC; HGB remains the frozen deployment estimator regardless of that comparison.

Metrics and inference. Primary metrics are mean NFE, exact task accuracy, elapsed time, and strict shadow risk. Every method receives the same prompt order and decode settings. We report paired bootstrap 95% intervals for NFE and accuracy differences (10,000 resamples), normalized cross-model NFE, simultaneous calibration bounds, and a failure taxonomy. The positive headline is allowed only if: (1) a non-mock certificate exists; (2) history-aware RC-PAG lowers NFE versus AdaBlock on both models and versus the best nonlearned rule overall; (3) each in-domain accuracy-difference lower bound versus AdaBlock is at least -0.02; and (4) the history-versus-local NFE interval excludes zero without higher empirical calibration risk.

Table 1: Frozen confirmatory decisions. These gates are evaluated mechanically before any positive headline is emitted.

Decision	Frozen rule
Risk target	$\alpha = 0.05$, familywise $\delta = 0.05$ over 12 model–policy pairs
Accuracy noninferiority	paired 95% lower bound ≥ -0.02
Compute	below AdaBlock on both models; below best heuristic overall
History contribution	paired NFE interval excludes 0; no higher calibration risk
Fallback	full refinement if no candidate is certified

Compute-aware funnel. We first run 32 prompts/model, then project A100-hours and storage. Only if the pilot is healthy do we collect 600 training traces/model, tune on 150 prompts/model, and calibrate on 300 prompts/model. Confirmation cannot start without an explicit flag, a certificate for both variants, and a calibration-time compute advantage over the selected nonlearned rule. All records are atomic and resumable; models are loaded one at a time on one A100.

5 Results

Confirmatory results pending. The build contains no generated numerical claims because a complete, non-mock confirmation directory was not supplied. The release script refuses mock artifacts. Table 1 records the analysis that will be inserted without changing the method or claim gates.

Regardless of outcome, the final report includes all candidates and all datasets. A failed certificate is itself a valid result and triggers the conservative fallback; a compute win with a failed accuracy gate is reported as a trade-off, not as an improvement.

6 Limitations and broader impact

The certificate is distribution- and protocol-specific. Dataset shift, a different model revision, stochastic temperature, altered block size, quantization, or a modified prompt template requires new calibration. Prompt-level shadow disagreement is stricter than many semantic metrics but does not guarantee correctness, factuality, fairness, or harmlessness. Conversely, a disagreement may be semantically benign. Calibration shadows increase experimental compute, although deployment does not pay that cost. Bonferroni correction is simple and robust but conservative, especially for correlated policies. The evaluation uses two open dLM families and reasoning/code tasks; it should not be generalized to all models, languages, or domains. Faster generation can lower energy per response but may also increase aggregate use. We release exact compute and failure records to make both effects measurable.

7 Conclusion

RC-PAG reframes adaptive diffusion decoding as constrained risk minimization. It combines compact phase-aware features with same-state counterfactual labels and a selection-safe finite-sample certificate. The result is a decoder that either chooses a lower-compute policy under an explicit premature-commitment risk target or falls back transparently. The frozen evaluation is designed so the strength of the paper follows the evidence rather than the other way around.

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A Proof details

Fix an unsafe candidate λ with $r = \mathcal{R}(\lambda) > \alpha$. Under the assumptions, $K_\lambda \sim \mathrm{Binomial}(n, r)$. For every integer k, the binomial lower-tail probability $\Pr_r(K_\lambda \leq k)$ is nonincreasing in r. Therefore the rejection region defined by $p_\lambda \leq \delta/M$, whose critical value is computed at α, has probability at most δ/M under every $r > \alpha$. By the union bound, the event that at least one of at most M unsafe candidates is certified has probability at most δ. On the complement, the certified set contains only safe policies; arbitrary selection by NFE within that set cannot invalidate the statement. If the set is empty, full refinement is returned and no statistical assertion about an adaptive candidate is made.

B Online algorithm

For each block, RC-PAG repeats: (1) execute one denoising step; (2) summarize the logits and token changes on device; (3) append the summary to a length-four history; (4) obtain $\widehat{r}_\theta$ from the CPU estimator; and (5) commit only after the candidate's minimum step and patience conditions hold. During calibration only, a commit also clones the pre-commit state, runs full refinement, and records Equation 2. Importantly, the shadow cannot change the live trajectory. The prompt loss is the maximum over its block comparisons.

C Full confirmatory results

Complete tables are generated only from a validated non-mock run.

D Reproducibility checklist

The artifact records repository and dataset revisions, the config and protocol hashes, prompt IDs, raw outputs, correctness, NFE, elapsed time, per-block compact traces, selected policy, shadow losses, calibration counts and p-values, paired intervals, and every failed claim gate. Code execution grading uses a separate subprocess with a timeout. Secrets are read from the environment and never written to the command line or artifacts. Exact local and Slurm commands are provided in `docs/rc_pag_runbook.md`.